

PROJECT MANAGEMENT

THE MANAGERIAL PROCESS

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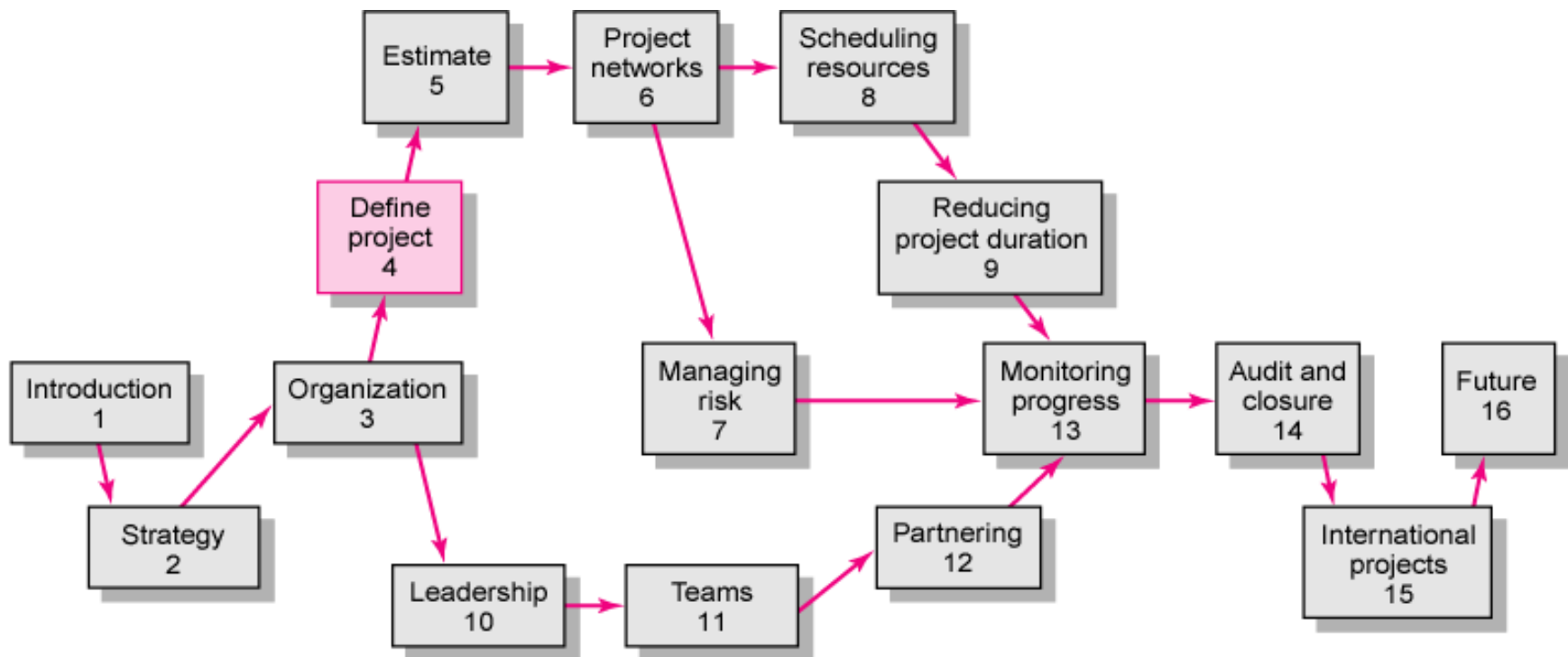
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Third Edition



Chapter 4

Defining the Project



Defining the Project

Step 1: Defining the Project Scope

Step 2: Establishing Project Priorities

Step 3: Creating the Work Breakdown Structure

Step 4: Integrating the WBS with the Organization

Step 5: Coding the WBS for the Information System

Step 1: Defining the Project Scope

- Project Scope

- A definition of the end result or mission of the project—a product or service for the client/customer—in specific, tangible, and measurable terms.

- Purpose of the Scope Statement

- To clearly define the deliverable(s) for the end user.
 - To focus the project on successful completion of its goals.
 - To be used by the project owner and participants as a planning tool and for measuring project success.

Project Scope Checklist

1. Project objective
2. Deliverables
3. Milestones
4. Technical requirements
5. Limits and exclusions
6. Reviews with customer



Project Scope: Terms and Definitions

- Scope Statements
 - Also called statements of work (SOW)
- Project Charter
 - Can contain an expanded version of scope statement
 - A document authorizing the project manager to initiate and lead the project.
- Project Creep
 - The tendency for the project scope to expand over time due to changing requirements, specifications, and priorities.

Step 2: Establishing Project Priorities

- Causes of Project Trade-offs
 - Shifts in the relative importance of criteria related to cost, time, and performance parameters
 - Budget–Cost
 - Schedule–Time
 - Performance–Scope
- Managing the Priorities of Project Trade-offs
 - Constrain: a parameter is a fixed requirement.
 - Enhance: optimizing a parameter over others.
 - Accept: reducing (or not meeting) a parameter requirement.

Project Management Trade-offs

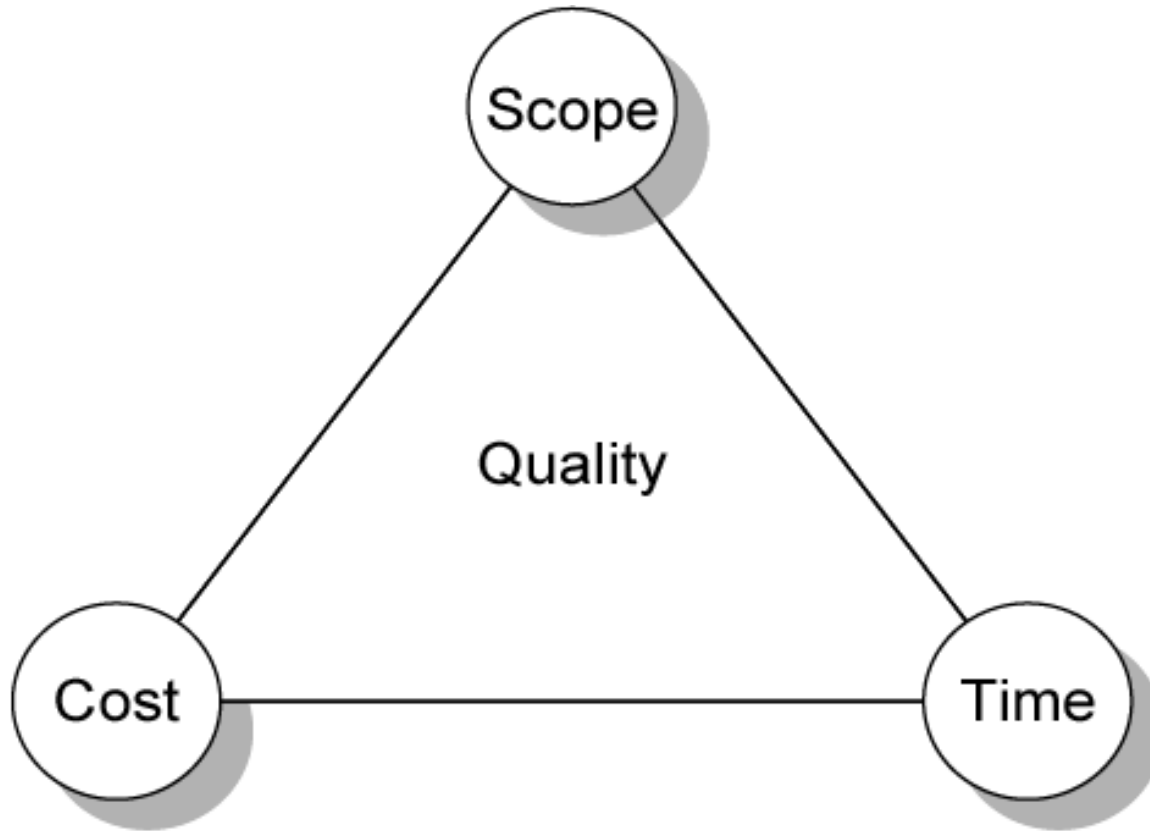


FIGURE 4.1

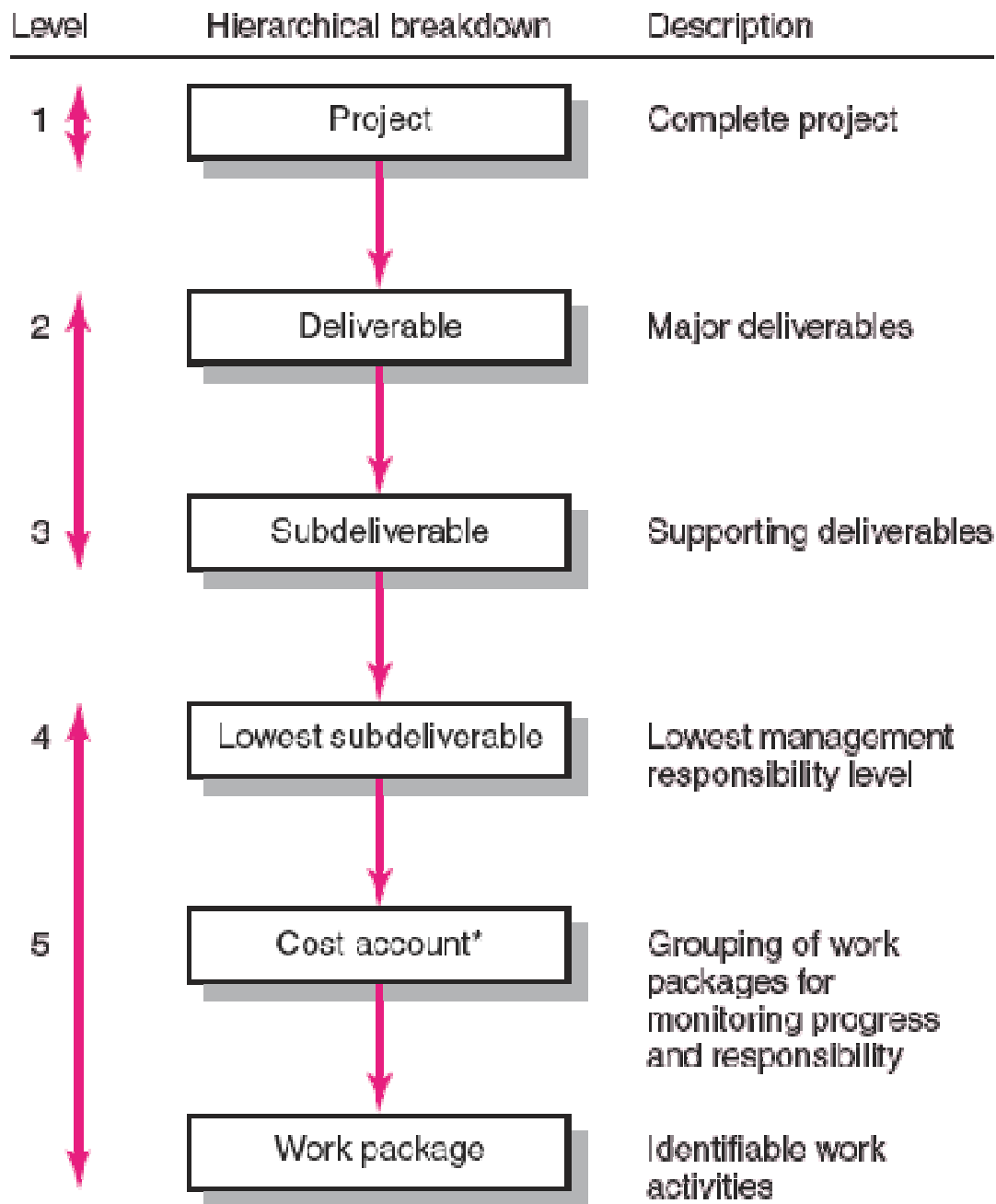
Project Priority Matrix

	Time	Performance	Cost
Constrain			
Enhance			
Accept			

FIGURE 4.2

Step 3: Creating the Work Breakdown Structure

- Work Breakdown Structure (WBS)
 - An hierarchical outline (map) that identifies the products and work elements involved in a project.
 - Defines the relationship of the final deliverable (the project) to its sub-deliverables, and in turn, their relationships to work packages.
 - Best suited for design and build projects that have tangible outcomes rather than process-oriented projects.



Hierarchical Breakdown of the WBS

FIGURE 4.3

How WBS Helps the Project Manager

- WBS

- Facilitates evaluation of cost, time, and technical performance of the organization on a project.
- Provides management with information appropriate to each organizational level.
- Helps in the development of the organization breakdown structure (OBS). which assigns project responsibilities to organizational units and individuals
- Helps manage plan, schedule, and budget.
- Defines communication channels and assists in coordinating the various project elements.

Work Breakdown Structure

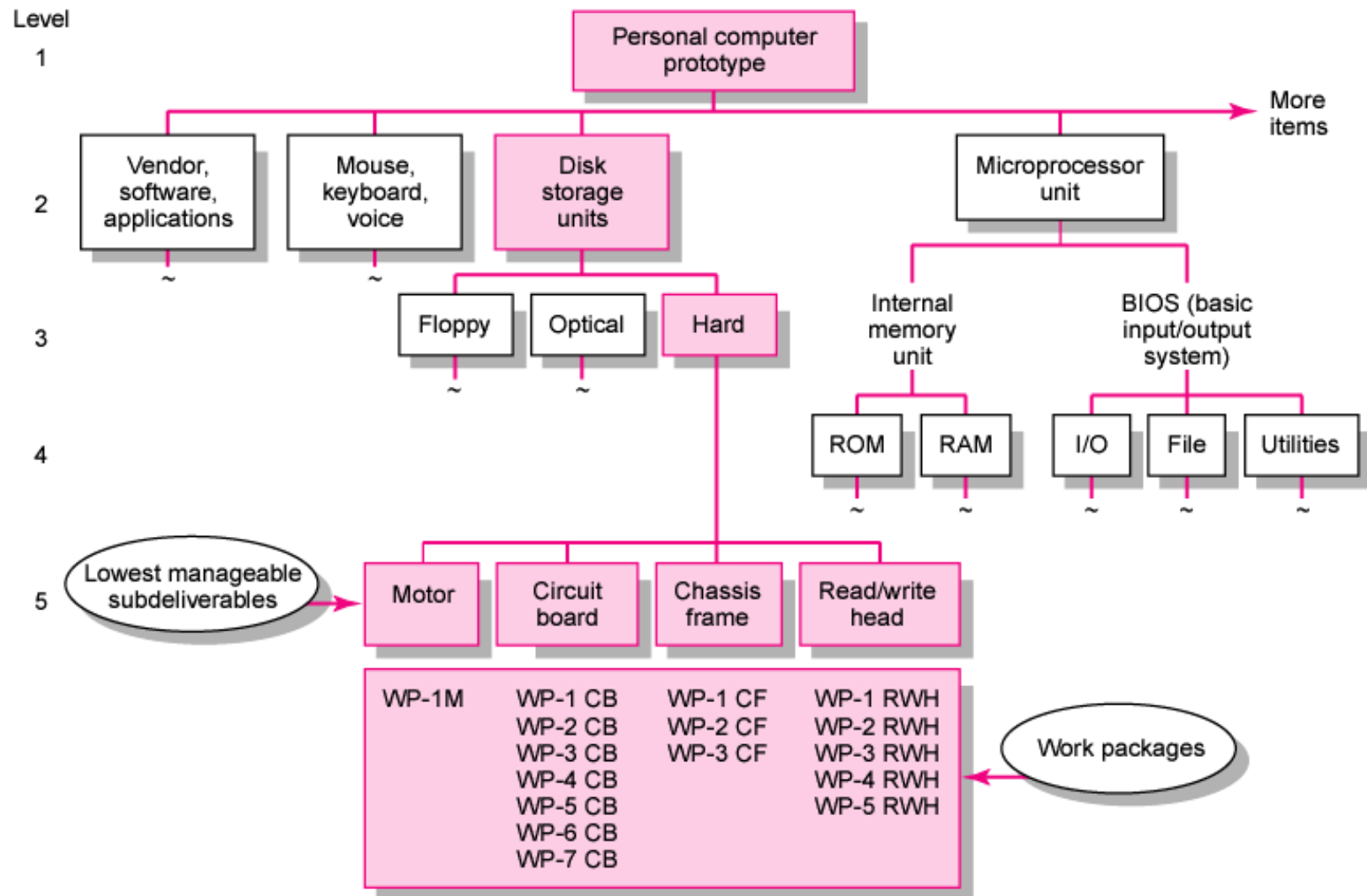


FIGURE 4.4

Work Packages

- A work package is the lowest level of the WBS.
 - It is output-oriented in that it:
 - Defines work (what).
 - Identifies time to complete a work package (how long)
 - Identifies a time-phased budget to complete a work package (cost)
 - Identifies resources needed to complete a work package (how much)
 - Identifies a single person responsible for units of work (who)
 - Identifies monitoring points (milestones) for measuring success.

Step 4: Integrating the WBS with the Organization

- Organizational Breakdown Structure (OBS)
 - Depicts how the firm is organized to discharge its work responsibility for a project.
 - Provides a framework to summarize organization work unit performance.
 - Identifies organization units responsible for work packages.
 - Ties the organizational units to cost control accounts.

Integration of WBS and OBS

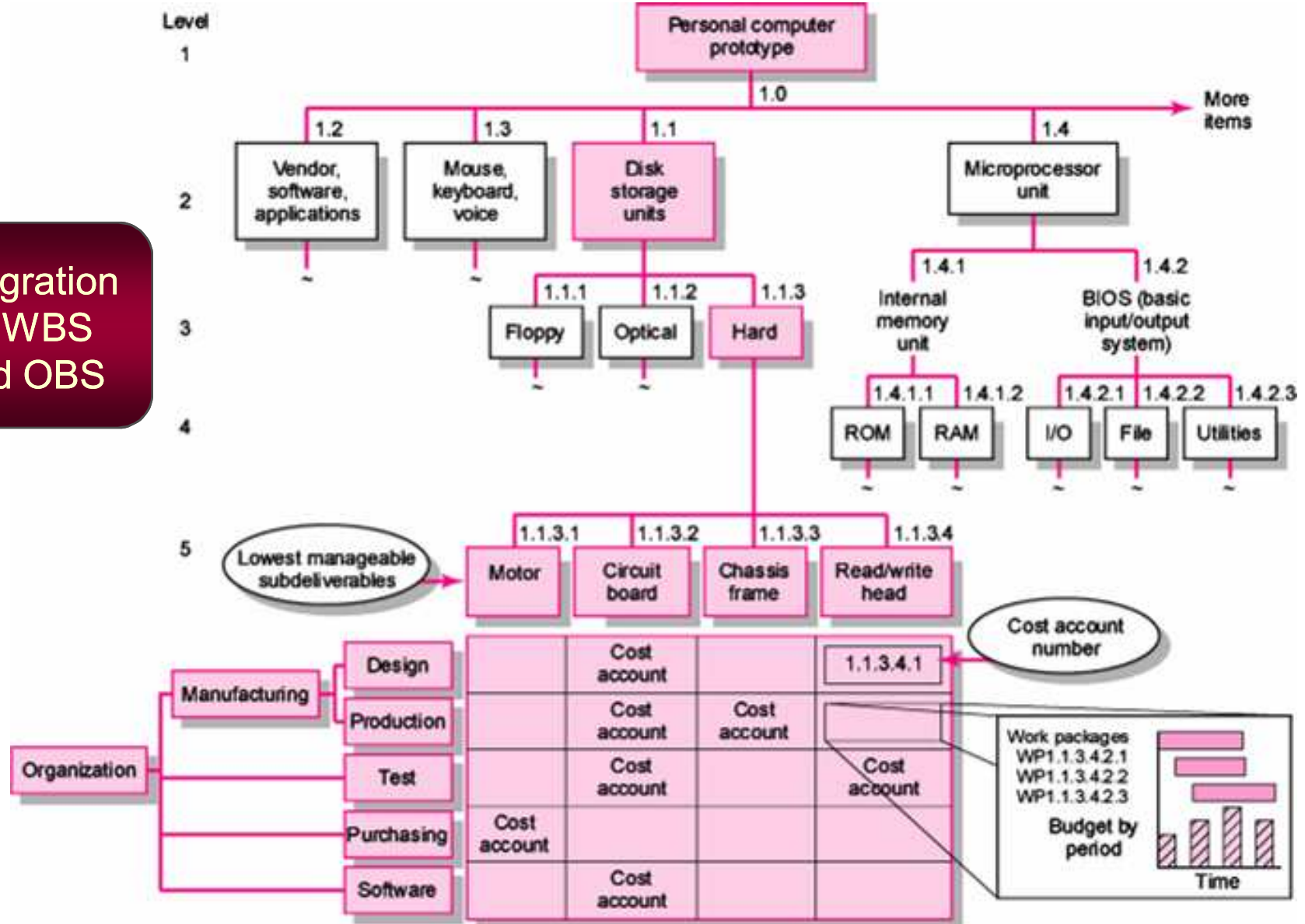


FIGURE 4.5

Direct Labor Budget Sorted By WBS

		Direct Labor Budget	
1.1.3	Hard drive	1,660	
1.1.3.1	Motor	10	
	Purchasing		10
1.1.3.2	Circuit board	1,000	
	Design		300
	Production		400
	Testing		120
	Software		180
1.1.3.3	Chassis frame	50	
	Production		50
1.1.3.4	Read/write head	600	
	Design		300
	Production		200
	Testing		100

TABLE 4.1A

Direct Labor Budget Sorted by OBS

Direct Labor Budget		
Design		600
1.1.3.2	Circuit board	300
1.1.3.4	Read/write head	300
Production		650
1.1.3.2	Circuit board	400
1.1.3.3	Chassis frame	50
1.1.3.4	Read/write head	200
Testing		220
1.1.3.2	Circuit board	120
1.1.3.4	Read/write head	100
Purchasing		10
1.1.3.1	Motor	10
Software		180
1.1.3.2	Circuit board	180
Total		1,660

TABLE 4.1B

Step 5: Coding the WBS for the Information System

- WBS Coding System

- Defines:

- Levels and elements of the WBS
 - Organization elements
 - Work packages
 - Budget and cost information

- Allows reports to be consolidated at any level in the organization structure



ID	Task Name
1	1 Computer project
2	1.1 Disk Storage units
3	1.1.1 Floppy
4	1.1.2 Optical
5	1.1.3 Hard
6	1.1.3.1 Motor
7	1.1.3.1.1 Sourcing work package
8	1.1.3.1.2*
9	1.1.3.1.3*
10	1.1.3.1.4*
11	1.1.3.2 Read/write head
12	1.1.3.2.1 Cost account
13	1.1.3.2.2 Cost account
14	1.1.3.2.3 WP
15	1.1.3.2.4 WP
16	1.1.3.2.5 WP
17	1.1.3.2.6 Cost account
18	1.1.3.2.7*
19	1.1.3.2.8*
20	1.1.3.2.9*

WBS Coding

Work Package Estimates

WP Description <u>Final version</u>	Page <u>1</u> of <u>1</u>
WP ID <u>1.1.3.2</u>	Project <u>PC proto</u>
Deliverable <u>Circuit board</u>	Date <u>9/29/XX</u>
Original Unit <u>Software</u>	Estimator <u>RMG</u>
WP Duration <u>3</u> work weeks	Total Budget \$ <u>265</u>

Time-Phased Budget (\$)

Work periods

Labor costs	Rate	1	2	3	4	5	Total
Code	\$ XX/hr	50	30	20			\$100
Document	\$ XX/hr		10	15			25
Publish	\$ XX/hr			5			5
Total labor		50	40	40			\$130
Materials			20				20
Equipment	\$ XX/hr	50	15	50			115
Other _____							
Total direct		100	75	90			\$265

FIGURE 4.6

Project Roll-up

- Cost Account

- The intersection of the WBS and the OBS that is a budgetary control point for work packages.
- Used to provide a roll-up (summation) of costs incurred over time by a work package across organization units and levels, and by deliverables.

Direct Labor Budget Rollup (000)

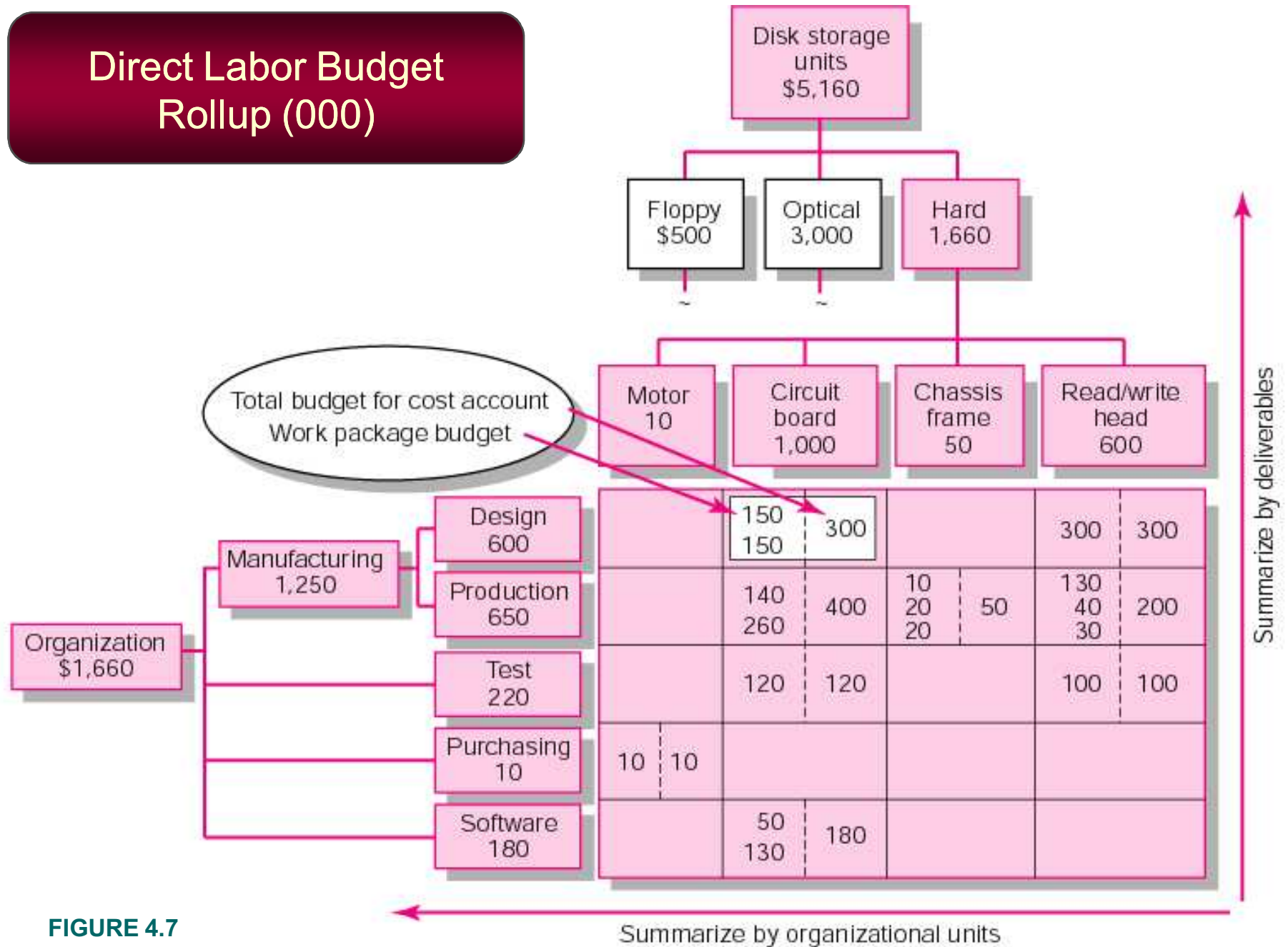


FIGURE 4.7

Process Breakdown Structure

- Process-Oriented Projects
 - Are driven by performance requirements in which the final outcome is the product of a series of steps of phases in which one phase affects the next phase.
- Process Breakdown Structure (PBS)
 - Defines deliverables as outputs required to move to the next phase .
 - Checklists for managing PBS:
 - Deliverables needed to exit one phase and begin the next.
 - Quality checkpoints for complete and accurate deliverables.
 - Sign-offs by responsible stakeholders to monitor progress.

PBS for Software Project Development

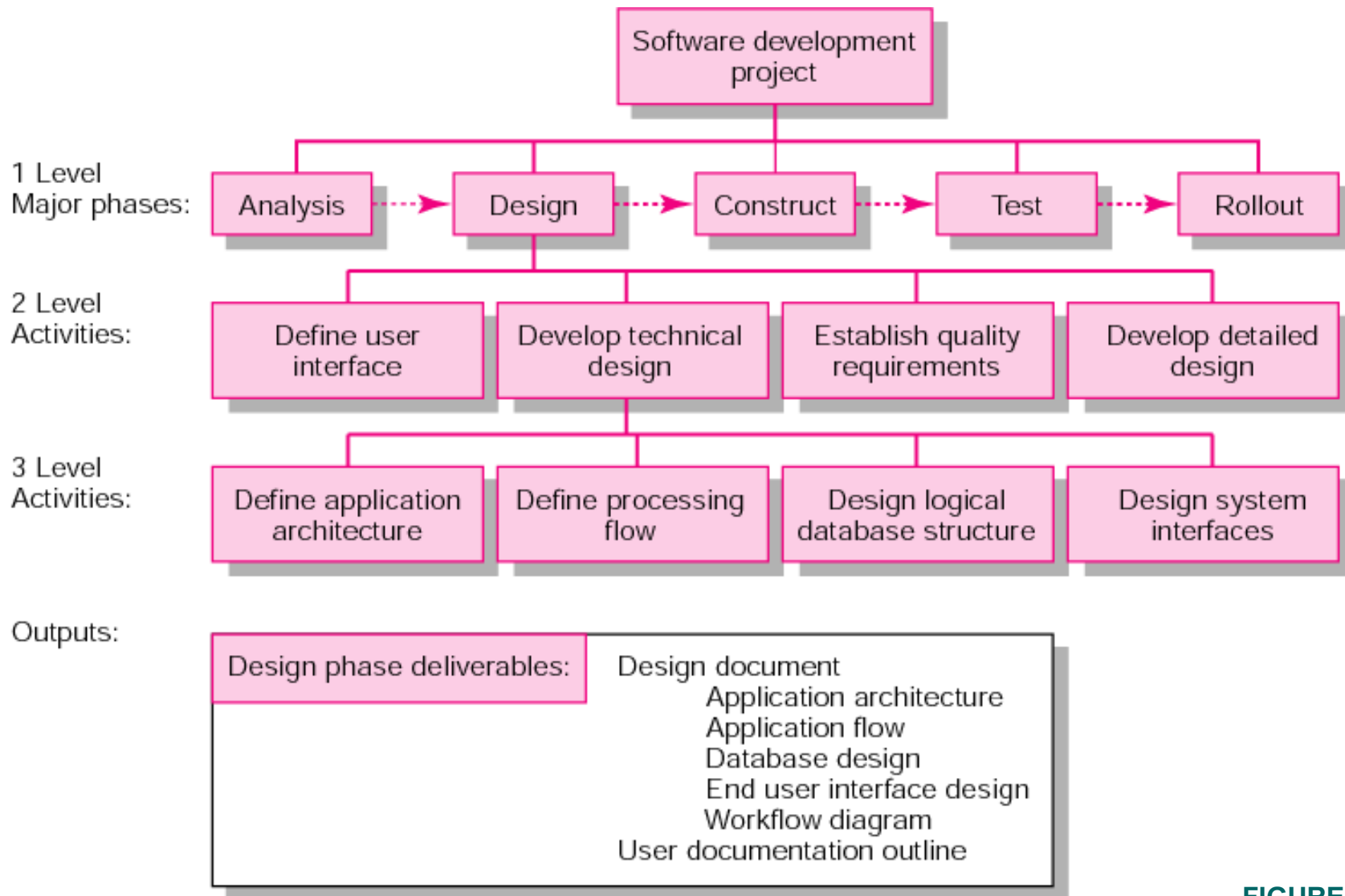


FIGURE 4.8

Responsibility Matrices

- Responsibility Matrix (RM)
 - Also called a linear responsibility chart.
 - Summarizes the tasks to be accomplished and who is responsible for what on the project.
 - Lists project activities and participants.
 - Clarifies critical interfaces between units and individuals that need coordination.
 - Provide an means for all participants to view their responsibilities and agree on their assignments.
 - Clarifies the extent or type of authority that can be exercised by each participant.

Responsibility Matrix for a Market Research Project

Task	Project Team				
	Richard	Dan	Dave	Linda	Elizabeth
Identify target customers	R	S		S	
Develop draft questionnaire	R	S	S		
Pilot-test questionnaire		R		S	
Finalize questionnaire	R	S	S	S	
Print questionnaire					R
Prepare mailing labels					R
Mail questionnaires					R
Receive and monitor returned questionnaires				R	S
Input response data			R		
Analyze results		R	S	S	
Prepare draft of report	S	R	S	S	
Prepare final report	R		S		

R = Responsible
S = Supports/assists

FIGURE 4.9

Responsibility Matrix for the Conveyor Belt Project

Deliverables	Organization							
	Design	Development	Documentation	Assembly	Testing	Purchasing	Quality Assur.	Manufacturing
Architectural design	1	2			2		3	3
Hardware specifications	2	1				2	3	
Kernel specifications	1	3						3
Utilities specification	2	1			3			
Hardware design	1			3		3		3
Disk drivers	3	1	2					
Memory management	1	3			3			
Operating system documentation	2	2	1					3
Prototypes	5		4	1	3	3	3	4
Integrated acceptance test	5	2	2		1		5	5

- 1 Responsible
- 2 Support
- 3 Consult
- 4 Notification
- 5 Approval

FIGURE 4.10

Key Terms

Cost account

Milestone

Organization breakdown structure (OBS)

Scope creep

Priority matrix

Responsibility matrix

Scope statement

Process breakdown structure (PBS)

Work breakdown structure (WBS)

Work package